Module 4: Numerical Methods - 1.

Numerical Solution of Algebraic and Transcendetal equations:-

Algebraic equations :- Equations involving algebraic quantities like x, x^2, x^3 are called as algebraic equations.

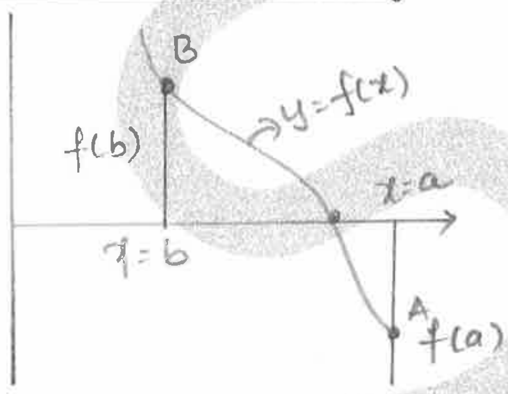
Eg: i) $x^4+x^3=80$ ii) $x^3-4x-9=0$.

Transcendental equations :- Equations involving non-algebraic quantities like e^x, $\log x$, $\sin x$, $\tan x$ etc are called as transcendental equations.

Eg: i) $xe^x - x = 0$ ii) $x\log_e x - 12 = 0$.

Iteration Process :- Numerical methods of finding approximate roots of the given equation is a repetative type of process known as iteration process.

Fundamental Property:-



If $f(x)=0$ is a real valued continuous function of the real variable x. We have the following fundamental property:
If there exists two values a, b such that $f(x)$ has opposite signs, say $f(a)<0$, $f(b)>0$ equivalently $f(a)f(b)<0$ then there always exists atleast one real root in the interval (a,b).

There are two numerical iterative methods:

i) Newton-Raphson Method.

ii) Regula-Falsi Method.

Newton-Raphson Method:

The first approximation to the root x_0 is given by $\boxed{x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}}$

The second approximation is obtained by replacing x_0 by x_1 in the RHS of the above expression.

In General, $\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}$ is called the Newton - Raphson iteration formula.

This procedure is repeated upto two consecutive approximated roots becomes equal.

Problems:-

1) Find a real root of the equation $x^3+x^2+3x+4=0$ by applying Newton-Raphson method. Carryout two iterations.

Soln:- Let $f(x) = x^3+x^2+3x+4$

$f(0)=4$, $f(-1)=1$, $f(-2)=-6<0$

$\therefore$ a real root lies in $(-2,-1)$

Let $x_0=-1$. Also $f'(x)=3x^2+2x+3$.

Ist Approximation: $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = -1 - \frac{f(-1)}{f'(-1)} = -1.25$

IInd Approximation: $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = -1.25 - \frac{f(-1.25)}{f'(-1.25)}$

$x_2 = -1.2229$.

$\therefore$ Thus the required real root is $-1.2229 \simeq -1.223$.

2) Use Newton-Raphson method to find a real root of the equation $x^3-2x-5=0$. Correct to three decimal Places.

Solⁿ :- Let $f(x)=x^3-2x-5$

$f(0)=-5<0$, $f(1)=-6<0$, $f(2)=-1<0$, $f(3)=16>0$

A root lies in $(2,3)$.

Let $x_0=2$, $f'(x)=3x^2-2$.

I^{st} Approximation: $x_1=x_0-\frac{f(x_0)}{f'(x_0)}=2-\frac{f(2)}{f'(2)}=2.1$.

II^{nd} Approximation: $x_2=x_1-\frac{f(x_1)}{f'(x_1)}=2.1-\frac{f(2.1)}{f'(2.1)}=2.0946$.

III^{rd} Approximation: $x_3=x_2-\frac{f(x_2)}{f'(x_2)}=2.0946-\frac{f(2.0946)}{f'(2.0946)}=2.0946$

Thus the required approximate root correct to three decimal places is 2.095.

3) Use Newton-Raphson method to find a real root of $x\sin x+\cos x=0$ near $x=\pi$. Carry out the iterations upto four decimal places of accuracy.

Solⁿ: Let $f(x)=x\sin x+\cos x$.

Let $x_0=\pi$, $f'(x)=x\cos x+\cancel{\sin x}-\cancel{\sin x}=x\cos x$.

$$x_1=x_0-\frac{f(x_0)}{f'(x_0)}=\pi-\frac{f(\pi)}{f'(\pi)}=\pi-\frac{(\pi\sin\pi+\cos\pi)}{\pi\cos\pi}=\pi-\frac{1}{\pi}=2.8233$$

$$x_2=x_1-\frac{f(x_1)}{f(x_2)}=2.7986\ ;\ x_3=x_2-\frac{f(x_2)}{f'(x_2)}=2.7984$$

$$x_4=x_3-\frac{f(x_3)}{f'(x_3)}=2.7984.$$

Thus the required real root is 2.7984.

4) Find a real root of the equation $xe^x - 2 = 0$ Correct to three decimal using Newton-Raphson method.

Soln:- Let $f(x) = xe^x - 2$.

$f(0) = -2 < 0$, $f(1) = 0.7183 > 0$

$\therefore$ a real root lies in $(0,1)$.

Let $x_0 = 1$, $f'(x) = xe^x + e^x = e^x(x+1)$.

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{f(1)}{f'(1)} = 1 - \frac{0.7183}{e'(2)} = 0.8679$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.8528 \; ; \; x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.8526$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.8526 .$$

Thus the required real root correct to three decimal places is 0.853.

5) Using Newton-Raphson Method find an approximate root of the equation $x \log_{10} x = 1.2$ correct to five decimal places that near 2.5

Soln :- Let $f(x) = x \log_{10} x - 1.2$, $x_0 = 2.5$

$$f(x) = x \cdot \frac{\log_e x}{\log_e 10} - 1.2$$

$$f'(x) = \frac{1}{\log_e 10}\left[1 + \log_e x\right]$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2.5 - \frac{f(2.5)}{f'(2.5)} = 2.7465$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.74065$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.74065.$$

$\therefore$ The required approximate root is 2.74065.

Regula Falsi Method or Method of False Position:

To find the numerical solution of the equation $f(x)=0$ by Regula-falsi method or method of false position.

Let the root of the eq^n $f(x)=0$ lies between a and b then $f(a)$ and $f(b)$ are of opposite sign.

$\therefore$ The first approximation of the root is given by

$$\boxed{x = \frac{a f(b) - b f(a)}{f(b) - f(a)}}$$

The procedure is repeated upto two consecutive approximate root becomes equal.

Problems:

1) Use the Regulafalsi method to find a real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places.

Sol^n :- Let $f(x) = x^3 - 2x - 5$

$f(0) = -5$, $f(1) = -6$, $f(2) = -1 < 0$, $f(3) = 16 > 0$.

$\therefore$ root lies in $(2, 3)$.

Since -1 is nearer to zero we can take the initial approximation. x_0 as nearer to 2.

Let $f(2.1) = 0.061 > 0$ $\quad \because f(2)$ is $-$ve & $f(2.1)$ is $+$ve

Roots lies b/n $(\underset{a}{2}, \underset{b}{2.1})$

I^{st} Approximation: $x_1 = \dfrac{af(b) - bf(a)}{f(b) - f(a)}$

$a = 2$; $f(a) = f(2) = -1$
$b = 2.1$; $f(b) = f(2.1) = 0.061$

$\therefore \quad x_1 = 2.0942$.

$f(x_1) = f(2.0942) = -0.00392 < 0$.
$\therefore$ Roots lies b/n $(\underset{a}{2.0942}, \underset{b}{2.1})$

II^{nd} Approximation: $x_2 = \dfrac{af(b) - bf(a)}{f(b) - f(a)}$

$a = 2.0942$; $f(a) = f(2.0942) = -0.00392$

$b = 2.1$; $f(b) = f(2.1) = 0.061$

$x_2 = 2.09455$.

Since upto 3 decimal places $x_1 = x_2 = 2.094$.

Thus the required approximate root is 2.094.

2) Find the real root of the equation $\tan x + \tanh x = 0$ lies between 2 and 3 by using regula falsi method. Carry out 3 iterations.

Sol^n:- Let $f(x) = \tan x + \tanh x$.
$f(2) = \tan 2 + \tanh 2 = -1.221 < 0$
$f(3) = \tan 3 + \tanh 3 = 0.8525 > 0$.

I^{st} Approximation: $x_1 = \dfrac{af(b) - bf(a)}{f(b) - f(a)}$.

$a = 2$, $f(a) = f(2) = -1.221$
$b = 3$, $f(b) = f(3) = 0.8525$
$x_1 = 2.59$
$f(x_1) = f(2.59) = 0.3735 > 0$.

$\therefore$ Roots lies in $(\underset{a}{2}, \underset{b}{2.59})$

II^{nd} Approximation :- $x_2 = \dfrac{af(b)-bf(a)}{f(b)-f(a)}$

$a=2$, $f(a)=f(2)=-1.221$

$b=2.59$, $f(b)=f(2.59)=0.3735$.

$x_2 = 2.4518$.

$f(x_2) = f(2.4518) = 0.1603>0$.

$\therefore$ Roots lies in $(\underset{a}{2}, \underset{b}{2.4518})$

III^{rd} Approximation :- $x_3 = \dfrac{af(b)-bf(a)}{f(b)-f(a)}$.

$a=2$, $f(a)=f(2)=-1.221$

$b=2.4518$, $f(b)=f(2.4518)=0.1603$

$x_3 = 2.3994$.

$\therefore$ Thus the required third approximation is 2.3994 .

3) Use the regula falsi method to obtain a root of the Equation $2x-\log_{10}x=7$ which lies b/n 3.5 and 4 .

Soln :- Let $f(x) = 2x-\log_{10}x-7$

$f(3.5) = -0.5441<0$, $f(4) = 0.3979>0$.

$\therefore$ Roots lies b/w $(3.5, 4)$

$a=3.5$ $\quad f(a)=f(3.5)=-0.5441$

$b=4$ $\quad f(b)=f(4)=0.3979$

I^{st} Approximation: $x_1 = \dfrac{af(b)-bf(a)}{f(b)-f(a)} = 3.79$

$f(x_1) = f(3.79) = 0.00136>0$.

$\therefore$ Root lies in $(3.5, 3.79)$

II^{nd} Approximation: $x_2 = \dfrac{a\,f(b) - b\,f(a)}{f(b) - f(a)}$.

$a = 3.5,\ f(a) = f(3.5) = -0.5441$

$b = 3.79,\ f(b) = f(3.79) = 0.00136$

$$x_2 = 3.7893 \simeq 3.79$$

Thus the approximate root correct to two decimal places is 3.79.

4) Find the smallest positive root of the equation $x^2 - \log_e x = 12$ by Regula-Falsi method.

Soln:- Let $f(x) = x^2 - \log_e x - 12$

$f(1) = -11 < 0$, $f(2) = -8.69 < 0$, $f(3) = -4.0986 < 0$.

$f(4) = 2.6137 > 0$.

$\therefore$ A real root lies in $(3, 4)$ and will be in the neighbourhood of 4.

$\therefore f(3.6) = -0.3209$, $f(3.7) = 0.3817$

$\therefore$ Roots lies in $(3.6, 3.7)$

I^{st} Approximation:- $x_1 = \dfrac{a\,f(b) - b\,f(a)}{f(b) - f(a)}$.

$a = 3.6,\ f(3.6) = -0.3209$; $b = 3.7,\ f(3.7) = 0.3817$

$$x_1 = 3.6457$$

$f(x_1) = -0.00242 < 0$.

$\therefore$ Roots lies in $(3.6457, 3.7)$

II^{nd} Approximation:- $x_2 = \dfrac{a\,f(b) - b\,f(a)}{f(b) - f(a)}$

$a = 3.6457,\ f(a) = -0.00242$; $b = 3.7$; $f(b) = 0.3817$

$$x_2 = 3.646042$$

Thus the required root correct to 3 decimal places is 3.646.

5) Find the root of the equation $xe^x - \cos x = 0$ by the method of false position.

Sol^n:- Let $f(x) = xe^x - \cos x$.

$f(0) = -1 < 0$, $f(1) = 2.178 > 0$

$f(0.5) = -0.0532 < 0$, $f(0.6) = 0.2679 > 0$

$\therefore$ Root lies in $(0.5, 0.6)$

Ist Approximation : $x_1 = \dfrac{a f(b) - b f(a)}{f(b) - f(a)}$

$a = 0.5$, $f(0.5) = -0.0532$

$b = 0.6$, $f(0.6) = 0.2679$

$x_1 = 0.5166$

$f(x_1) = f(0.5166) = -0.0035 < 0$

$\therefore$ Root lies in $(0.5166, 0.6)$

IInd Approximation : $x_2 = \dfrac{a f(b) - b f(a)}{f(b) - f(a)}$

$a = 0.5166$, $f(0.5166) = -0.0035$

$b = 0.6$ $f(0.6) = 0.2679$

$x_2 = 0.5177$

$f(x_2) = f(0.5177) = -0.00017 < 0$

$\therefore$ Root lies in $(0.5177, 0.6)$

IIIrd Approximation : $x_3 = \dfrac{a f(b) - b f(a)}{f(b) - f(a)}$

$a = 0.5177$, $f(a) = -0.00017$

$b = 0.6$, $f(b) = 0.2679$

$x_3 = 0.5178$.

Thus the required root is 0.5178.

Finite Differences:

Let $y = f(x)$ be a function, in this function 'x' is independent variable and 'y' is dependent Variable. By assigning a value for x, we get a corresponding value $y = f(x)$ for the function. The value assigned to x is called the argument. The corresponding value of the function is called entry. The difference between 2 consecutive values of x is called the interval of difference. It is denoted by 'h'.

Forward Difference:

Let $y = f(x)$ be a function & 'h' is the interval of difference, then $f(x+h) - f(x)$ is called the first forward difference of the function between the arguments x and $x+h$. It is denoted by $\Delta f(x)$.

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta f(x_0) = f(x_0+h) - f(x_0) \; ; \; \Delta y_0 = y_1 - y_0$$

$$\Delta f(x_1) = f(x_1+h) - f(x_1) \; ; \; \Delta y_1 = y_2 - y_1$$

$$\vdots$$

$$\Delta f(x_n) = f(x_n+h) - f(x_n) = \Delta y_{n-1} = y_n - y_{n-1}$$

The difference of the first forward differences are called second forward differences.

i.e. $\Delta^2 y_0 = \Delta y_1 - \Delta y_0$.

$$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$$

$$\vdots$$

$$\Delta^2 y_{n-2} = \Delta y_{n-1} - \Delta y_{n-2}.$$

Forward Difference Table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	---	$\Delta^n y$
x_0	y_0					
		Δy_0				
x_1	y_1		$\Delta^2 y_0$			
		Δy_1		$\Delta^3 y_0$		
x_2	y_2		$\Delta^2 y_1$			
		Δy_2	$\vdots$	$\vdots$	----	$\Delta^n y_0$.
x_3	y_3					
$\vdots$	$\vdots$	$\vdots$		$\Delta^3 y_{n-3}$		
			$\Delta^2 y_{n-2}$			
x_{n-1}	y_{n-1}					
		Δy_{n-1}				
x_n	y_n					

The first entries in the table $\Delta y_0, \Delta^2 y_0, \Delta^3 y_0$ ---- $\Delta^n y_n$ are called the leading forward differences.

Backward Difference:

Let $f(x)$ be a given function & 'h' be the difference of interval, then the backward difference ∇ (nabla) is defined as $\nabla f(x) = f(x) - f(x-h)$

$$\nabla f(x_n) = f(x_n) - f(x_n - h) \; ; \; \nabla y_n = y_n - y_{n-1}$$

$$\nabla f(x_{n-1}) = f(x_{n-1}) - f(x_{n-1} - h) \; ; \; \nabla y_{n-1} = y_{n-1} - y_{n-2}$$

$$\vdots$$

$$\nabla y_2 = y_2 - y_1$$

$$\nabla y_1 = y_1 - y_0$$

Similarly second backward difference are defined by

$$\nabla^2 y_n = \nabla y_n - \nabla y_{n-1}$$

$$\nabla^2 y_{n-1} = \nabla y_{n-1} - \nabla y_{n-2}$$

$$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$$

Backward difference table:

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	---	$\nabla^n y$
x_0	y_0					
x_1	y_1	∇y_1	$\nabla^2 y_2$			
x_2	y_2	∇y_2		$\nabla^3 y_2$		
x_3	y_3	¦	¦	¦	---	$\nabla^n y_n$
¦	¦	¦	¦	$\nabla^3 y_n$		
¦	¦	¦	$\nabla^2 y_n$			
x_{n-1}	y_{n-1}	∇y_n				
x_n	y_n					

The last entries in the table $\nabla y_n, \nabla^2 y_n, \nabla^3 y_n$ --- $\nabla^n y_n$ are called the leading backward differences.

Interpolation:-

The method of finding the value of 'y' at a specified value of 'x', having known values of x and y is called Interpolation.

Interpolation when the values of 'x' are equally spaced.

* Newton-Gregory forward interpolation formula :-

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \text{-----}$$

Where $p = \dfrac{x - x_0}{h}$.

$x \rightarrow$ The value of x at which y is to be calculated.

$x_0 \rightarrow$ Initial value of 'x'

$h \rightarrow$ Common difference / Step length.

$\Delta \rightarrow$ Forward difference operator.

Newton Gregory backward interpolation formula :-

$$y = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \cdots$$

Where $p = \frac{x - x_n}{h}$

$x \rightarrow$ The value of x at which y is to be calculated

$x_n \rightarrow$ final value of 'x'

$h \rightarrow$ Common difference / step length.

$\nabla \rightarrow$ Backward difference operator.

Note :-

* If the value of 'x' is near beginning of the table, then we use Newton's forward interpolation formula.
* If the value of 'x' is near end of the table, then we use Newton's backward interpolation formula.

Problems :-

1) Find $y(0.12)$ given the data

x	0.10	0.15	0.20	0.25	0.30
y	0.1003	0.1511	0.2027	0.2553	0.3033

Sol^n: Since $x = 0.12$ is close to first value of x, hence we have to apply Newton forward interpolation formula.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.10	0.1003				
		0.0508			
0.15	0.1511		0.0008		
		0.0516		0.0002	
0.20	0.2027		0.0010		0.0002
		0.0526		0.0004	
0.25	0.2553		0.0014		
		0.054			
0.30	0.3093				

We have NGFIF,

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0$$

$$p = \frac{x - x_0}{h} = \frac{0.12 - 0.10}{0.05} = 0.4 .$$

$$\boxed{y(0.12) = 0.12054}$$

2) Using Newtons forward interpolation formula find an interpolating polynomial, that approximate the function given by the following table.

x	0	1	2	3	4
y	-5	-10	-9	4	35

Hence find $y(0.5)$.

Soln :-

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	-5				
		-5			
1	-10		6		
		1		6	
2	-9		12		0
		13		6	
3	4		18		
		31			
4	35				

We have NFIF,

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \cdots$$

$$p = \frac{x - x_0}{h} = \frac{x-0}{1} = x.$$

$$\boxed{y = x^3 - 6x - 5.}$$

At $x = 0.5$ $\boxed{y(0.5) = -7.875}$

3) The area of a circle (A) corresponding to diameter (D) is given below.

D	80	85	90	95	100
A	5026	5674	6362	7088	7854

Find the area corresponding to diameter 105 using an appropriate interpolation formula.

Sol^n:- Since $x = 105$ is close to end value of x, hence we have to apply Newton's backward interpolation formula.

$x = D$	$y = A$	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
80	5026				
		648			
85	5674		40		
		688		-2	
90	6362		38		4
		726		2	
95	7088		40		
		766			
100	7854				

By NBIF.

$$y = y_n + P\nabla y_n + \frac{P(P+1)}{2!}\nabla^2 y_n + \frac{P(P+1)(P+2)}{3!}\nabla^3 y_n + \frac{P(P+1)(P+2)(P+3)}{4!}\nabla^4 y_n$$

$$P = \frac{x - x_n}{h} = \frac{105-100}{5} = 1$$

$$\therefore \boxed{y(105) = 8666.}$$

4) From the following table find the number of students who have obtained a) less than 45 marks
b) between 40 and 45 marks
c) less than 55 marks.

Marks	30-40	40-50	50-60	60-70	70-80
No of students	31	42	51	35	31

Soln:- Let $y = f(x)$ denote the no of students who have obtained $\leq$ 'x' marks.

Less than 40 marks = 31
Less than 50 marks = 31+42 = 73
Less than 60 marks = 73+51 = 124
Less than 70 marks = 124+35 = 159
Less than 80 marks = 159+31 = 190.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
40	31				
		42			
50	73		9		
		51		-25	
60	124		-16		37
		35		12	
70	159		-4		
		31			
80	190				

i) ≤ 45 marks.

By NFIF.

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0.$$

$$p = \frac{x - x_0}{h} = \frac{45-40}{10} = 0.5$$

$$\boxed{y(45) = 47.867 \simeq 48 \text{ students.}}$$

ii) between 40 & 45 students.

Total no of students with 45 marks = 48.

Total no of students with ≤ 40 marks = 31

$\boxed{17}$.

iii) ≤ 55 marks.

By NFIF,

$$p = \frac{55-40}{10} = 1.5$$

$$\boxed{y(55) = 99.8 \simeq 100 \text{ students.}}$$

6) A Survey Conducted in a slum locality reveals the following information as classified below.

Income per day (Rs)	Under 10	10-20	20-30	30-40	40-50
No of persons	20	45	115	210	115

Estimate the probable number of persons in the income group of 20 to 25

Soln: By NFIF, $p = \frac{x - x_0}{h} = \frac{25-10}{10} = 1.5$

$y(25) = 107$.

From table $y(20) = 65$. $\therefore y(25) - y(20) = \underline{\underline{42}}$

6) The population for a town is given by the following table.

Year	1951	1961	1971	1981	1991
Population in thousands	19.96	39.65	58.81	77.21	94.61

Using Newton's forward and backward interpolation formula, calculate the increase in the population from the year 1955 to 1985.

Solⁿ: i) To find 1955.

By NFIF, $P = \frac{x - x_0}{h} = \frac{1955 - 1951}{10} = 0.4$.

$\boxed{y(1955) = 27.89}$.

ii) To find 1985

By NBIF, $P = \frac{x - x_n}{h} = \frac{1985 - 1991}{10} = -0.6$

$\boxed{y(1985) = 84.3}$.

Thus the increase in population from the year 1955 to 1985 is given by $84.3 - 27.89 = 56.41$ thousands.

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7) Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ = 0.8660$, find $\sin 57^\circ$ using an appropriate interpolation formula.

Solⁿ:- Since 57° is near end value of x.

By NBIF,

$$P = \frac{x - x_n}{h} = \frac{57 - 60}{5} = -0.6 .$$

$\therefore \boxed{y(57) = \sin 57^\circ = 0.8387}$.

Interpolation when the values of 'x' are not equally spaced :

* Divided Difference :—

Let $f(x)$ be the given function $x_0, x_1, x_2 --- x_n$ are the Values of x, which are not equally spaced.

Let $f(x_0), f(x_1) ---- f(x_n)$ are the values of function at the point $x_0, x_1, x_2 --- x_n$ respectively.

The first divided difference between the arguments x_0 & x_1 is denoted & defined as :

$$f(x_0, x_1) = \Delta f(x_0) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}.$$

$$f(x_1, x_2) = \Delta f(x_1) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\vdots$$

$$f(x_{n-1}, x_n) = \Delta f(x_{n-1}) = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

The second divided difference is denoted and defined as $f(x_0, x_1, x_2) = \Delta^2 f(x_0) = \dfrac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}.$

Newton divided difference formula:

Newton general interpolation formula is given by

$$y = f(x) = f(x_0) + (x - x_0)\Delta f(x_0) + (x - x_0)(x - x_1)\Delta^2 f(x_0)$$
$$+ (x - x_0)(x - x_1)(x - x_2)\Delta^3 f(x_0) + - - - - -$$

($\Delta \rightarrow$ Divided Difference operator)

Problems :-

1) Given the values

x :	5	7	11	13	17
$f(x)$:	150	392	1452	2366	5202

Using divided difference formula for unequal intervals, to find $f(9)$.

Sol^n :

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
$5 = x_0$	150	$\frac{392-150}{7-5} = 121$		
$7 = x_1$	392	$\frac{1452-392}{11-7} = 265$	$\frac{265-121}{11-5} = 24$	$\frac{32-24}{13-5} = 1$
$11 = x_2$	1452	$\frac{2366-1452}{13-11} = 457$	$\frac{457-265}{13-7} = 32$	$\frac{42-32}{17-7} = 1$
$13 = x_3$	2366	$\frac{5202-2366}{17-13} = 709$	$\frac{709-457}{17-11} = 42$	
$17 = x_4$	5202			

We have NDDF.

$$y = f(x_0) + (x-x_0)\Delta f(x_0) + (x-x_0)(x-x_1)\Delta^2 f(x_0) + (x-x_0)(x-x_1)(x-x_2)\Delta^3 f(x_0).$$

$$y = 150 + (9-5)\times 121 + (9-5)(9-7)\times 24 + (9-5)(9-7)(9-11)\times 1$$

$$\boxed{y(9) = f(9) = 810.}$$

2) Using Newton's divided difference interpolation formula to find $f(8)$ and $f(15)$ from given data

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028.

Sol^n:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
4	48			
		$\frac{100-48}{5-4}=52$		
5	100		$\frac{97-52}{7-4}=15$	
		$\frac{294-100}{7-5}=97$		$\frac{21-15}{10-4}=1$
7	294		$\frac{202-97}{10-5}=21$	
		$\frac{900-294}{10-7}=202$		$\frac{27-21}{11-5}=1$
10	900		$\frac{310-202}{11-7}=27$	
		$\frac{1210-900}{11-10}=310$		$\frac{33-27}{13-7}=1$
11	1210		$\frac{409-310}{13-10}=33$	
		$\frac{2028-1210}{13-11}=409$		
13	2028			

By NDDF,

i) $f(8) = 448$; ii) $f(15) = 3150$. [Polynomial $f(x) = x^3 - x^2$]

3) Construct the interpolation polynomial for the data given below using Newton's general interpolation formula for divided differences.

x	2	4	5	6	8	10
y	10	96	196	350	868	1746

Sol^n:

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
2	10			
		$\frac{96-10}{4-2}=43$		
4	96		$\frac{100-43}{5-2}=19$	
		$\frac{196-96}{5-4}=100$		$\frac{27-19}{6-2}=2$
5	196		$\frac{154-100}{6-4}=27$	
		$\frac{350-196}{6-5}=154$		$\frac{35-27}{8-4}=2$
6	350		$\frac{259-154}{8-5}=35$	
		$\frac{868-350}{8-6}=259$		$\frac{45-35}{10-5}=2$
8	868		$\frac{439-259}{10-6}=45$	
		$\frac{1746-868}{10-8}=439$		
10	1746			

By NDDF,

$$f(x) = f(x_0) + (x-x_0)\Delta f(x_0) + (x-x_0)(x-x_1)\Delta^2 f(x_0) + (x-x_0)(x-x_1)(x-x_2)\Delta^3 f(x_0) + \cdots$$

$$f(x) = 10 + (x-2)(43) + (x-2)(x-4)(19) + (x-2)(x-4)(x-5)(2)$$

$$\boxed{f(x) = 2x^3 - 3x^2 + 5x - 4}$$

4) Use Newton's divided difference formula to find $f(4)$ given the data:

x	0	2	3	6
$f(x)$	-4	2	14	158.

Ans: $f(4) = 40$.

* Lagrange's Interpolation Formula:

Let $y = f(x)$ be a function $y_0 = f(x_0)$, $y_1 = f(x_1) \dots y_n = f(x_n)$ are the values of the function for the arguments $x_0, x_1 \dots x_n$ which are unequally spaced.

The lagrange's interpolation formula is given by

$$y = f(x) = \frac{(x-x_1)(x-x_2)\cdots(x-x_n)}{(x_0-x_1)(x_0-x_2)\cdots(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2)\cdots(x-x_n)}{(x_1-x_0)(x_1-x_2)\cdots(x_1-x_n)} y_1 + \frac{(x-x_0)(x-x_1)\cdots(x-x_n)}{(x_2-x_0)(x_2-x_1)\cdots(x_2-x_n)} y_2 + \cdots + \frac{(x-x_0)(x-x_1)\cdots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\cdots(x_n-x_{n-1})} y_n.$$

* Note: i) $(x-a)(x-b) = x^2 - (a+b)x + ab$.

ii) $(x-a)(x-b)(x-c) = x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc$.

Problems :-

1) Use Lagrange's interpolation formula to fit a polynomial for the data.

	x_0	x_1	x_2	x_3
x	0	1	3	4
y	-12	0	6	12
	y_0	y_1	y_2	y_3

Hence estimate y at $x=2$.

Soln:- We have Lagrange's interpolation formula,

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1 +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$y = f(x) = \frac{(x-1)(x-3)(x-4)}{(-1)(-3)(-4)}(-12) + 0 + \frac{x(x-1)(x-4)}{(3)(2)(-1)}(6) +$$

$$\frac{x(x-1)(x-3)}{(4)(3)(1)}(12)$$

$$f(x) = (x-1)(x-3)(x-4) - x(x-1)(x-4) + x(x-1)(x-3)$$

$$\boxed{y = f(x) = x^3 - 7x^2 + 18x - 12.}$$

At $x=2$, $\boxed{y = 4.}$

2) Use Lagrange's interpolation formula to find y at $x=10$ given,

x	5	6	9	11
y	12	13	14	16

Sol^n: $y = f(x) = \dfrac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0 + \dfrac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

$$f(10) = \frac{(4)(1)(-1)}{(-1)(-4)(-6)}(12) + \frac{(5)(1)(-1)}{(1)(-3)(-5)}(13) + \frac{(5)(4)(-1)}{(4)(3)(-2)}(14) +$$

$$\frac{(5)(4)(1)}{(6)(5)(2)} 16$$

$\boxed{f(10) = 14.67}$ at $x = 10$.

3) Use Lagrange's interpolation formula to find $f(4)$ given

x	0	2	3	6
$f(x)$	-4	2	14	158

Sol^n: $f(4) = 40$.

4) Find $y(9)$ using Lagrange's Interpolation formula for the given data

x	5	7	11	13	17
y	150	392	1452	2366	5202

Sol^n:- $y(9) = 810.006$.

5) Find $y(11)$ using Lagrange's Interpolation formula for the given data

x	2	5	8	14
y	94.8	87.9	81.8	68.7

Sol^n:- $y(11) = 74.925$

Numerical Integration:

This is the process of obtaining approximate value of the definite integral $I = \int_a^b y\,dx$ without actually integrating the function, but only using values of 'y' at some points of 'x' equally spaced over $[a, b]$.

There are 3 rules to obtain the value of the definite integral numerically.

Ist Rule :— Trapezoidal Rule:

$$\int_{x_0}^{x_0+nh} y\,dx = \frac{h}{2}\left[y_0 + 2(y_1 + y_2 + y_3 + \cdots + y_{n-1}) + y_n\right]$$

IInd Rule :- Simpson's 1/3rd Rule:

$$\int_{x_0}^{x_0+nh} y\,dx = \frac{h}{3}\left[(y_0 + y_n) + 4(y_1 + y_3 + \cdots + y_{n-1}) + 2(y_2 + y_4 + \cdots y_{n-2})\right]$$

1st last   4 (odd)   2 (even)

IIIrd Rule :- Simpson's 3/8th Rule:

$$\int_{x_0}^{x_0+nh} y\,dx = \frac{3h}{8}\left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \cdots) + 2(y_3 + y_6 + \cdots y_{n-3})\right]$$

1st last   3 (non multiples of 3)   2 (Multiples of 3)

Note:—

* $h \rightarrow$ Common difference.
* Here 'n' is the total number of sub intervals of the interval.
* If there are 'n' intervals then there are $(n+1)$ no of ordinates.

Problems :–

1) Evaluate $\int_2^7 \frac{dx}{x}$ using Trapezoidal rule, taking $n=5$.

Sol^n:- $x_0 = 2$, $x_0 + nh = 7$, $n = 5$

$2 + 5h = 7$

$5h = 5$

$h = 1.$

x	2	3	4	5	6	7
$y = \frac{1}{x}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$
	y_0	y_1	y_2	y_3	y_4	y_5

By Trapezoidal rule.

$$\int_{x_0}^{x_0+nh} y\,dx = \frac{h}{2}\left[(y_0 + y_5) + 2(y_1 + y_2 + y_3 + y_4)\right]$$

$$\int_2^7 \frac{dx}{x} = \frac{1}{2}\left[\left(\frac{1}{2} + \frac{1}{7}\right) + 2\left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)\right]$$

$$\boxed{\int_2^7 \frac{dx}{x} = 1.2714}$$

2) Use the Trapezoidal Rule with 6 subintervals to approximate $\int_0^2 \frac{dx}{16+x^2}$.

Sol^n:- $x_0 = 0$, $x_0 + nh = 2$, $n = 6$.

$0 + 6h = 2$

$h = 2/6 = 1/3$.

x	0	$\frac{1}{3}$	$\frac{2}{3}$	1	$\frac{4}{3}$	$\frac{5}{3}$	2
$y = \frac{1}{16+x^2}$	0.0625	0.062	0.0608	0.0588	0.0562	0.0532	0.05
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

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By Trapezoidal rule,

$$\int_{x_0}^{x_0+nh} y\,dx = \frac{h}{2}\left[(y_0+y_6)+2(y_1+y_2+y_3+y_4+y_5)\right]$$

$$\int_0^2 \frac{dx}{16+x^2} = \frac{1/3}{2}\left[(0.0625+0.05)+2(0.062+0.0608+0.0588+0.0562+0.0532)\right]$$

$$\boxed{\int_0^2 \frac{dx}{16+x^2} = 0.1158.}$$

3) Find solution using Trapezoidal rule

x	1.4	1.6	1.8	2.0	2.2
$f(x)$	4.0552	4.9530	6.0436	7.3891	9.0250

Soln: $\int_{x_0}^{x_0+nh} y\,dx = 4.9852.$

4) Evaluate $I = \int_0^1 \frac{1}{1+x^2}\,dx$ by Simpson's $1/3^{rd}$ rule. By considering 6 subintervals of the interval $[0,1]$. Hence find an approximate value of π.

Soln:- $x_0 = 0$, $x_0+nh = 1$, $n = 6$.

$0+6h = 1$

$h = 1/6$.

x	0	$1/6$	$2/6$	$3/6$	$4/6$	$5/6$	$6/6 = 1$
$y = \frac{1}{1+x^2}$	1	0.9730	0.9	0.8	0.6923	0.5901	0.5
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By Simpson's $\frac{1}{3}^{rd}$ rule,

$$\int_0^1 \frac{1}{1+x^2}\,dx = \frac{h}{3}\left[(y_0+y_6)+4(y_1+y_3+y_5)+2(y_2+y_4)\right]$$

$$\boxed{\int_0^1 \frac{1}{1+x^2}\,dx = 0.7854}\,.$$

<u>To find π:</u>

We have $\int_0^1 \frac{1}{1+x^2}\,dx = 0.7854$.

$$\tan^{-1}x\Big]_0^1 = 0.7854$$

$$\tan^{-1}1 - \cancel{\tan^{-1}0} = 0.7854$$

$$\frac{\pi}{4} = 0.7854$$

$$\boxed{\pi = 3.1416.}$$

5) Evaluate $I = \int_0^5 \frac{1}{4x+5}\,dx$ by Simpson's $\frac{1}{3}^{rd}$ rule by considering 10 sub intervals of the interval. Hence find an approximate value of $\log 5$.

Sol^n:- $x_0=0$, $x_0+nh=5$, $n=10$.

$0+10h=5$

$h=\frac{1}{2}$.

x	0	$\frac{1}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{4}{2}$	$\frac{5}{2}$	$\frac{6}{2}$	$\frac{7}{2}$	$\frac{8}{2}$	$\frac{9}{2}$	$\frac{10}{2}$
y	0.2	0.1428	0.1111	0.0909	0.0769	0.0666	0.0588	0.0526	0.0476	0.0434	0.04

By Simpson's $\frac{1}{3}^{rd}$ Rule,

$$\int_0^5 \frac{1}{4x+5}\,dx = \frac{h}{3}\Big[(y_0+y_{10})+4(y_1+y_3+y_5+y_7+y_9)+2(y_2+y_4+y_6+y_8)\Big]$$

$$\boxed{\int_0^5 \frac{1}{4x+5}\,dx = 0.4023}$$

To find log5:

$$\int_0^5 \frac{1}{4x+5}\,dx = 0.4023.$$

$$\frac{1}{4}\log(4x+5)\Big|_0^5 = 0.4023.$$

$$\frac{1}{4}[\log(25)-\log(5)] = 0.4023.$$

$$\log\left(\frac{25}{5}\right) = 1.6092$$

$$\boxed{\log 5 = 1.6092}$$

6) Evaluate $\int_0^{\pi/2} \cos x\,dx$ by applying Simpson's $\frac{1}{3}^{rd}$ rule by taking eleven ordinates.

Soln:- $x_0 = 0$, $x_0 + nh = \pi/2$

$0 + 10h = \frac{\pi}{2}$

$h = \frac{\pi}{20} = 9^\circ.$

11- ordinates.

10+1 - Ordinates

$n+1 \Rightarrow n = 10$ (Sub intervals)

x°	0°	9°	18°	27°	36°	45°	54°	63°	72°	81°	90°
$y = \cos x$	1	0.9877	0.9511	0.8910	0.8090	0.7071	0.5878	0.454	0.3090	0.1564	0
	y_0	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}

By Simpson's $\frac{1}{3}^{rd}$ Rule.

$$\int_a^b y\,dx = \frac{h}{3}\left[(y_0+y_{10})+4(y_1+y_3+y_5+y_7+y_9)+2(y_2+y_4+y_6+y_8)\right]$$

$$\therefore \boxed{\int_0^{\pi/2} \cos x\,dx = 1.}$$

Here $h = \frac{\pi}{20}$

$\pi = 22/7$.

7) Using Simpson's $\frac{3}{8}^{th}$ rule evaluate $I = \int_0^1 \frac{x}{1+x^2}\,dx$ with $n=6$. Hence find an approximate value of i) $\log 2$ ii) $\log\sqrt{2}$.

Soln:- $x_0 = 0$, $x_0 + nh = 1$, $n = 6$

$6h = 1$

$h = 1/6$.

x	0	$1/6$	$2/6$	$3/6$	$4/6$	$5/6$	$6/6 = 1$
$y = \frac{x}{1+x^2}$	0	0.1621	0.3	0.4	0.4615	0.4918	0.5
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

By Simpson's $\frac{3}{8}^{th}$ rule.

$$\int_{x_0}^{x_0+nh} y\,dx = \int_0^1 \frac{x}{1+x^2}\,dx = \frac{3h}{8}\left[(y_0+y_6)+3(y_1+y_2+y_4+y_5) + 2(y_3)\right]$$

$$\boxed{\int_0^1 \frac{x}{1+x^2}\,dx = 0.3467.}$$

To find $\log 2$:

$$\int_0^1 \frac{x}{1+x^2}\,dx = 0.3467.$$

$$\frac{1}{2}\int_0^1 \frac{2x}{1+x^2}\,dx = 0.3467$$

$\frac{1}{2} \log(1+x^2)\Big]_0^1 = 0.3467$.

$\log 2 - \log_0 1 = 0.6934$.

$\boxed{\log 2 = 0.6934.}$

To find $\log \sqrt{2}$:

$\int_0^1 \frac{x}{1+x^2} dx = 0.3467$; $\frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx = 0.3467$.

$\frac{1}{2} (\log 2 - \log 1) = 0.3467$

$\boxed{\log \sqrt{2} = 0.3467.}$

8) Use Simpson's $\frac{1}{3}^{rd}$ rule to find $\int_0^{0.6} e^{-x^2} dx$ by taking 6 sub-intervals.

Sol^n^: $\int_0^{0.6} e^{-x^2} dx = 0.5351.$